DP-PostMerge: Free Compression via Post-hoc Merging for Multi-Vector Visual Document Retrieval

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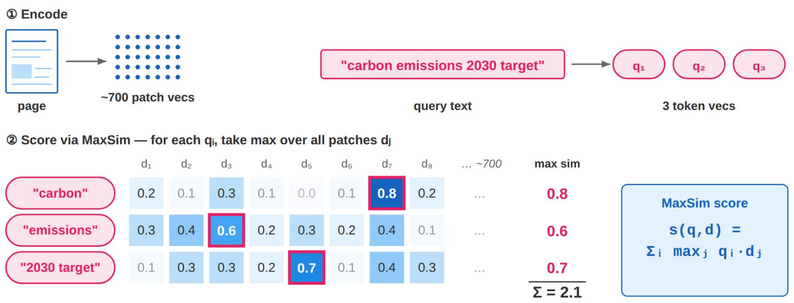
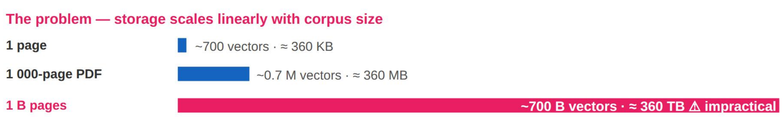
PROBLEM · MOTIVATION · METHOD

Problem

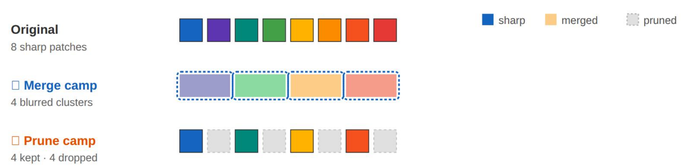
Visual Document Retrieval (VDR). SOTA models like **ColPali / ColQwen** encode each page as ~700 patch vectors, then score queries via **MaxSim late interaction**:





Motivation

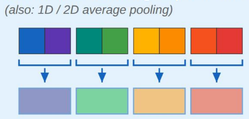
Two research camps disagree on how to compress:



Merge camp — *Light-ColPali · Ma 2025*

"Pruning is inherently unsuitable."

Hierarchical clustering on patch embeddings
(also: 1D / 2D average pooling)



4 cluster centroids (averaged colors)

Why no pruning?
Their tests showed fixed-ratio pruning collapses past 50% — judged unsalvageable.

✓ Graceful at high comp · ✗ blurs signals

Prune camp — *DocPruner · Yan 2025*

"You pruned wrong — fixed-ratio is naive."

Per-doc adaptive attention threshold
τ = μ + k·σ on EOS attention scores

4 kept (attn > τ), 4 dropped

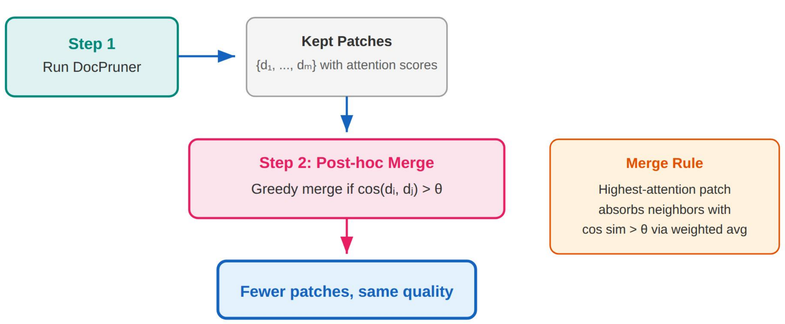
Why no merging?
Averaging patches into centroids dilutes the sharp signals MaxSim relies on.

✓ Sharp at low/mid comp · ✗ collapses >80%

Our take — adaptive prune, then merge surviving dupes

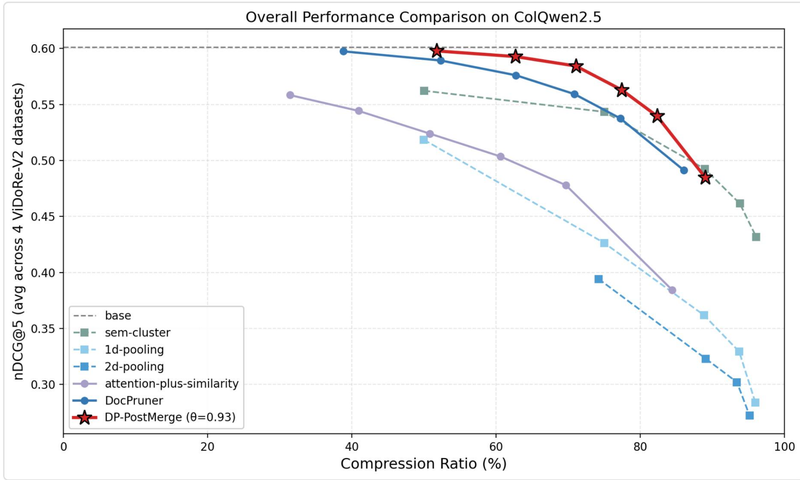
8 → 5 sharp → 4 vectors · same budget, no signal blur, no near-duplicates

Method



MAIN FINDING — VIDORE-V2 (4 DATASETS, COLQWEN2.5)

Quantitative results : ViDoRe-V2



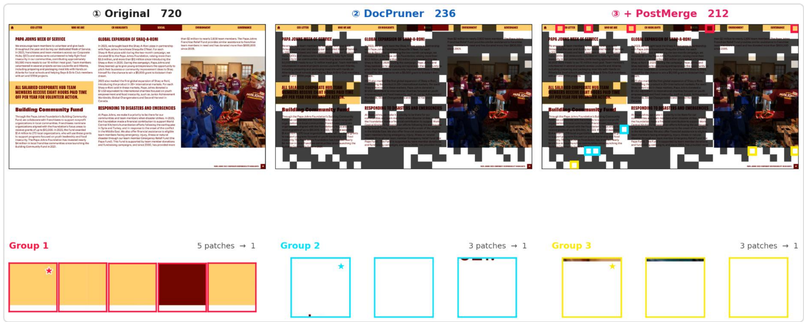
nDCG@5 vs compression on ViDoRe-V2 · avg of 4 datasets · 6 baselines + DP-PostMerge (★)

DP-PostMerge Pareto-dominates all baselines from 50% to 80% compression

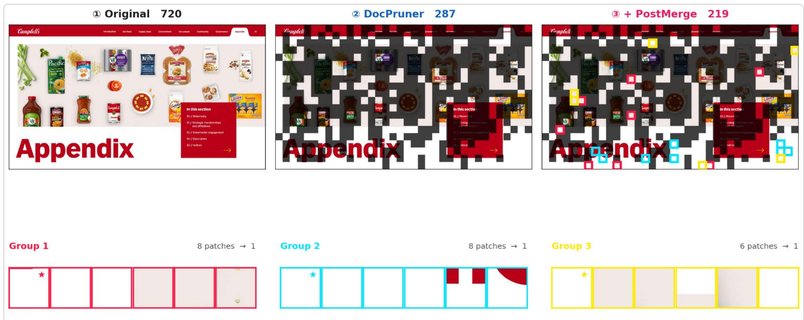
k	DocPruner nDCG@5	DocPruner Compress.	DP-PostMerge nDCG@5	DP-PostMerge Compress.	Free Δ
−0.5	**0.598**	**~39%**	**0.598**	**~52%**	**+13%**
−0.25	0.589	~52%	**0.593**	~63%	**+10%**
0.00	**0.582**	**~63%**	**0.584**	**~71%**	**+8%**
0.25	0.559	~71%	**0.563**	~77%	**+7%**
0.50	**0.537**	**~77%**	**0.540**	**~82%**	**+5%**

*Same DocPruner k, ± post-hoc merge (θ = 0.93). PM gives **+3–13% storage for free** at matched (or better) quality.*

Qualitative results : What Gets Merged



***Example 1 — ESG report cover** · 720 → 236 → 212 vectors · grey = pruned, color = merge group (★ anchor)*

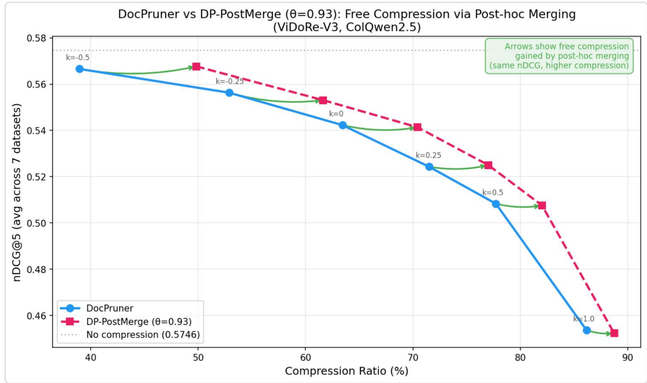


***Example 2 — ESG appendix** · 720 → 287 → 219 vectors · clusters: white tiles, red tiles, food tiles*

What both examples reveal: the surviving "near-duplicate" groups are exactly the kinds of repeated content DocPruner can't distinguish — uniform backgrounds, repeated logos, structural padding, and tiled product images. Each absorbs **3–8 patches into one anchor vector** with no measurable retrieval loss, which is the source of DP-PostMerge's **3–13% extra free compression** in the table above.

ANALYSIS & GENERALIZATION

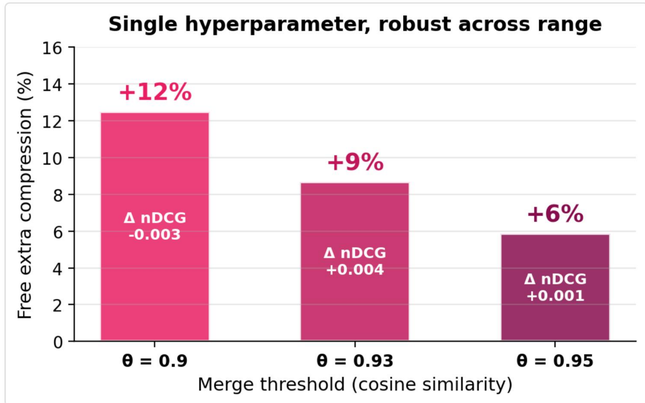
Generalization — ViDoRe-V3



nDCG@5 vs compression ratio (avg across 7 ViDoRe-V3 datasets). Same pattern holds on a larger, more diverse benchmark.

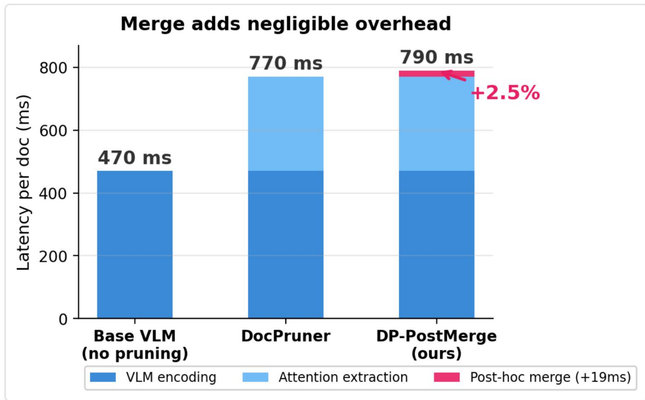
Same gains on 7 new datasets

θ Sensitivity



Avg across 4 ViDoRe-V2 datasets × 5 DocPruner k values. Δ nDCG = change vs DocPruner alone.

Runtime



Per-document latency on ColQwen2.5 (ViDoRe-V2, 1538 docs). VLM encoding dominates; merge adds only 19 ms (+2.5%).

Takeaways

1. **Importance ≠ redundancy.** DocPruner is near-optimal for selecting patches, but leaves exploitable redundancy among survivors.
2. **Free compression.** Post-hoc merge gives +3–10% extra compression at every operating point, with negligible quality loss and +2.5% runtime.
3. **Plug-and-play.** Zero training, single θ, backbone-agnostic — attaches after any pruner as a drop-in second stage.

Setup: ColQwen2.5 · ViDoRe-V2 (4 ds) + V3 (7 ds) · nDCG@5 · k ∈ {−0.5,…,1.0} · θ = 0.93 | **Refs:** DocPruner — Yan et al., 2025; ColPali — Faysse et al., 2024; ViDoRe — Macé et al., 2025 | **Code:** github.com/jianxyou/project_6765